

JOSEPH FRIMPONG

Postdoctoral Researcher

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EDUCATION

PhD in Chemistry, Wayne State University

Sept 2019 - Aug 2024

Major: Physical Chemistry - GPA-3.93

Detroit, MI

PhD Thesis: *Development & use of Computational Methods for the Electronic Structure of Quantum Materials and Interfaces*

Bachelor of Science in Chemistry, Kwame Nkrumah University of Science and Technology

Sept 2013 - July 2017

Major: Chemistry (Honors) GPA - 3.79

Ghana

SKILLS

Modelling Software

Quantum ESPRESSO, AMBER, TeraChem, Gaussian, BerkeleyGW, GROMACS, PySCF, WEST

Programming Languages

Python, Bash Scripting, C++, FORTRAN, MATLAB, R, GIT

High-Performance Computing

Perlmutter, ACCESS (Bridges & Expanse), BNL, ANL (Carbon), WSU Grid, MPI, GPU programming

Visualization

Avogadro, VMD, ASE, GnuPlot, GaussView, Matplotlib, Blender

Data & Machine Learning

NumPy, Pandas, PyTorch, Matminer, DeepChem, Pymatgen, Tableau, Horovod, W&B, DDP

Instrumentation

UV-VIS, GC-MS, LC-MS, TLC, HPLC, IR, Ion Chromatograph, IoT [ESP32, Arduino, STM32]

WORK/RESEARCH EXPERIENCE

Postdoctoral Researcher - AI/ML for electronic structure calculations

Nov 2024 - Present

Argonne National Laboratory | Advisors: Prof. Maria Chan & Prof. Giulia Galli

Lemont, IL

- Conduct research on the integration of machine learning techniques with electronic structure calculations to accelerate computations of electronic and optical properties.
- Leveraging expertise in density functional theory (DFT) & GW+BSE, high-performance computing (HPC), and machine learning to address complex computational materials problems.
- Contributing to the development of MICCoM-supported codes for exascale simulations, focusing on algorithm optimization and parallel programming for CPU and GPU architectures.
- Using state-of-the-art HPC resources at Argonne to perform large-scale simulations and data analysis.

Schaap-Rumble Research Fellow | Graduate Research Assistant

Sept 2020 - July 2024

Wayne State University | Advisor: Prof. Zhenfei Liu

Detroit, MI

- Developed a GW/BSE method for studying large heterogeneous interfaces with **programming tools** like (MPI, FORTRAN, Python, BASH) and **implemented machine learning models** for materials discovery
- Employed ab-initio **GW/BSE** and Density functional theory (**DFT**) to understand and predict electronic energy level alignment and optical properties of Quantum Dots, & Covalent Organic Frameworks and organic-metal interfaces
- Collaborated with experimentalists on various projects including the selective isotopic labeling of drug molecules using Quantum dots towards the development of stable pharmaceutical drugs
- Performed **QM/MM calculations** to understand the effect of solvents in quantum dots and small molecules on the electronic and optical properties
- Created custom scripts tailored to streamline research workflows, resulting in improved data analysis and visualization capabilities
- Applied advanced **data analysis** and **graphical visualization** techniques to extract valuable insights into complex structure-property relationships, facilitating informed decision-making in materials design and optimization
- Utilized high-performance computing (HPC) resources for high-throughput screening of materials properties with SLURM and PBS job schedulers and **training of machine learning models on multiple GPUs**.
- Compiled CPU and GPU based simulation libraries efficiently on HPC compute clusters

Data Analyst (Project)

Sept 2022 – May 2023

Wayne State University

Detroit, MI

- Developed a **deep learning** model for predicting air quality using data from a personally custom-built ESP32 powered sensors
- Conducted **exploratory data analysis** on NIH proposals (big data) using Python (**Pandas and SQL**) presenting data in an interactive **Tableau** dashboard
- Built an **ML model** that tokenizes data to generate keywords given text data and a simple natural language processing model to answer questions about the data

Research Assistant

May 2016 - Aug 2018

Kwame Nkrumah University of Science and Technology | Advisors: Dr. Caroline Kwawu & Prof. Evans Adei

Kumasi, GH

- Employed DFT calculations using Quantum ESPRESSO to investigate catalytic mechanisms of CO₂ reduction on low Miller index iron surfaces (100), (010), and (111), enabling prediction of product selectivity

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- Elucidated the reaction mechanism of ruthenium-catalyzed cycloaddition between alkynes and strained cyclic alkenes using Gaussian and Spartan molecular modelling packages

Physics Teacher, *Tamale Senior High School*

Nov 2018 – Aug 2019

- Taught, assessed, and graded 210 high school students in general chemistry and physics. Improved the percentage pass rate from 50 to 86% in the annual exam
- Demonstrated, supervised, and developed new student-centered laboratory sessions in chemistry and physics

Teaching Assistant, *Kwame Nkrumah University of Science and Technology*

Sept 2017 – Aug 2018

- Taught experimental inorganic chemistry courses with 100% pass rate
- Taught students the technical know how of UV, MS, IR and HPLC instruments to perform molecular structural elucidation

PUBLICATIONS

6. Rajendra Maity, Otto Dungan, Frédéric A. Perras, Jingwei Li, Daohua Liu, Sumei Ren, Dan Lehnher, Zheng Huang, Eric M. Phillips, Moses Adeyemo, **Joseph Frimpong**, Timothy Quainoo, Zhen-Fei Liu, and Long Luo (2024), *Hydrogen Isotope Labeling of Pharmaceuticals Via Dual Hydrogen Isotope Exchange Pathways Using CdS Quantum Dot Photocatalyst*, JACS 146 (49), 34141-34151 [\[link\]](#)
5. **Frimpong* J.**, Aryal* S., Karun T., and Liu Z.(2024) *Understanding the ligand exchange in CdS quantum dots from first principles* Chem. Mater., 36, 23, 11519–11528 [\[link\]](#)
4. **Frimpong J.** and Liu Z. (2023) *Extending the substrate screening GW approximation to covalently bound interfaces* J. Phys. Chem. Lett. 2024, 15, 2133–2141 [\[link\]](#)
3. Behera N., Gunasekera D., Mahajan J., **Frimpong J.**, Liu Z, Luo L. (2023) *Electrochemical Hydrogen Isotope Exchange of Amines Controlled by Alternating Current Frequency*, Faraday Discuss.,247, 45-58 [\[link\]](#)
2. Aryal S*, **Frimpong J.*** and Liu Z.(2022) *Comparative Study of Covalent and van der Waals CdS Quantum Dot Assemblies from Many-Body Perturbation Theory*, J. Phys. Chem. Lett., 13, 43, 10153–10161 [\[link\]](#)
1. **Frimpong J.** and Liu Z. (2021) “*Quasiparticle electronic structure of two-dimensional heterotriangulene-based covalent organic frameworks adsorbed on Au(111)*” J. Phys.: Condens. Matter **33**, 254004 [\[link\]](#)

AWARDS AND HONORS

11. Better Scientific Software Honorable Mention Award (2026)
10. Center for Nanoscale Materials Grant for Computational work in Machine learning for electronic structure
9. Dan Trivich Memorial Award for Research in Physical Chemistry [**awarded to the best students in the Physical Chemistry Division**] (2024)
8. NOBCChE Advancing Science Conference Award (2022 & 2023)
7. Summer Dissertation Award @ Wayne State University [**awarded to top PhD candidates with exceptional projects**] (2023)
6. Energy Research Travel Funding (GERA), APS March Meeting, (Chicago, IL 2022 & Las Vegas, NV, 2023)
5. A. Paul and Carole C. Schaap Endowed Distinguished Graduate Award [**given to top graduate students in Chemistry**] (2022)
4. Thomas C. Rumble Fellowship Award [**in recognition of superior academic achievement in research**] (2022)
3. Best Poster Award — Computational Chemistry, NOBCChE National Conference (2022)
2. ICTP Travel Award — The Abdus Salam International Centre for Theoretical Physics, Italy (2019)
1. KBN Merit Scholarship Kwame Nkrumah University of Science and Technology, Ghana (2019)

TECHNICAL TALKS & PRESENTATIONS

10. **J. Frimpong** (2025) Towards Accelerating Excited State Calculations with Generative Modeling, Argonne Postdoctoral Research & Career Symposium, Nov 26, Detroit, MI
9. **J. Frimpong**, J. Zhan, Victor Yu, Guilia Galli, Maria Chan (2025) Towards Accelerating Excited State Calculations with Generative Modeling, Midwest Theoretical Chemistry Conference, May 26, Detroit, MI
8. **J. Frimpong** and Z. Liu (2024) Understanding ligand exchange in Quantum dots from first principles, ACS Spring Meeting, March 19, New Orleans, LA
7. **J. Frimpong** and Z. Liu (2023) Development of GW-based substrate screening methods for covalently-bound interfaces, NOBCChE National Conference 2023, September 12, New Orleans, LA

6. **J. Frimpong**, and Z Liu (2023) Development of GW-based substrate screening methods for strongly coupled molecule-metal interfaces, APS March Meeting 2023, March 18, Las Vegas, NV
5. **J. Frimpong**, S. Aryal, and Z. Liu (2022) Unraveling the role of Surface Ligands in the Electronic and Optical Properties of Cadmium Sulphide Quantum Dots, NOBCChE National Conference 2022, September 25, Orlando, FL
4. **J. Frimpong**, S. Aryal, and Z. Liu (2022) Electronic Structure and Optical Properties of Quantum Dots their Assemblies, APS March Meeting 2022, March 18, Chicago, IL
3. **J. Frimpong** and Z. Liu (2021) Electronic structure of 2D-covalent organic frameworks and the influence of Au(111) substrate, Chemistry Graduate Research Symposium (CGRS), Detroit, MI. (Student Talk)
2. **J. Frimpong** and Z. Liu (2021) Quasiparticle electronic structure of two-dimensional heterotriangulene-based covalent organic frameworks adsorbed on Au(111), APS March Meeting 2021, March 17 (Virtual)
1. **J. Frimpong** (2019) Mechanistic studies of the cycloaddition of norbornadiene and alkynyl phosphonates, 19th International Workshop on Computational Physics and Material Science, Trieste, Italy (Poster Presentation)

EXPERT TRAININGS AND SUMMER SCHOOLS

- Argonne Training Program on Extreme-Scale Computing (ATPESC) (July–Aug 2025)
- Deep Learning for Science Summer School (LBNL), Berkeley, CA, USA (June 2025)
- URSSI Summer School in Open Science and Research Software Engineering at UIUC (July–Aug 2024)

MENTORING EXPERIENCE

Amber Witherspoon , ACTSO '26	2025-2026
– Machine learning for the out-of-domain classification of Brain-Tumors)	
Tejas Karun, BS Chemistry WSU '23	2022-2023
– Designed and supervised project: Electronic Structure studies of 2DPA-1 using Density Functional Theory Studies (DFT)	
Lyric Elliott , BS Chemistry WSU '21	2021
– Designed and supervised project: Electronic Structure studies of quinoidal polymers using DFT	

LEADERSHIP/VOLUNTEER ACTIVITIES

- Professional Development Officer (Jan - Dec 2026) Postdoctoral Society of Argonne, Lemont, IL, USA
- **Social Media Coordinator** (Aug 2022 - July 2023) WSU NOBCChE Chapter, Detroit, MI, USA
- **NOBCChE's K-12 Initiative**(2021-2023) Detroit Public School, Detroit, MI, USA
- **NOBCChE Conference Planning Committee**(2022 – Present)
- **Facilitator** - Chemistry Meets Computers - Coding Summer Camp (2021, 2022, 2023) Wayne State University, Detroit, MI, USA
- **General Secretary** (Aug 2021 - July 2022) WSU NOBCChE Chapter
- **Volunteer** (Oct 2021, Nov 2023) Detroit Public School Outreach & **Volunteer** (March 2023) Neinas Chemistry Outreach,
- **Peer Reviewer** for Journal of Physics: Condensed Matter, Physical Review B, Physical Review Letters, IOP Physica Scripta, SGCS DOE Fellowship, ACS Journals,

PROFESSIONAL MEMBERSHIP

American Physical Society (APS) | American Chemical Society (ACS) | National Organization for the Professional Advancement of Black Chemists and Chemical Engineers (NOBCChE) | Sigma Xi Honor Society

REFERENCES

- Available on request.